CONCEPT NOTE · PRECISION AGRONOMY FOR WEST AFRICA

Growa AI

Translating high-resolution soil & weather telemetry into native-dialect farming advice.

The Problem

Agriculture is over 22% of Nigeria's real GDP, yet smallholder farmers face a 45% average yield gap. Low-cost IoT sensors now make precision soil and weather data widely available, but it arrives as raw numbers in technical English — useless to the many farmers who are non-literate or speak local dialects.

- 38M+ active smallholder farmers in Nigeria form the addressable base.
- A 45% yield gap restricts national agricultural GDP growth.
- Existing agri-tech tools assume literacy and English fluency, excluding most end users.

The Solution

Growa AI ingests soil and weather sensor data, runs it through a crop-yield prediction engine, and converts the result into plain-language advisory in English and West African Pidgin — delivered via app, SMS, and interactive voice response (IVR). It bridges high-tech sensing infrastructure and on-the-ground decisions for farmers who cannot read a soil report.

38M+ — smallholder farmers | 45% — avg. yield gap | $22B — maize/yam/cassava output

Product Overview

At the core of Growa AI is an interactive prediction tool. A farmer or extension agent picks a crop and location and enters two field-measurable inputs — soil pH and recent rainfall — and the engine returns a full agronomic picture in seconds.

Inputs

- Crop: Maize, Cassava, or Yam
- Location: Ogbomoso, Kaduna, or Enugu (regional climate models)
- Soil pH and recent rainfall (mm), measured on-site or via IoT probes

Outputs

- Yield probability (%) and a Low / Moderate / High risk rating, with the primary risk issue (e.g. drought, flooding, acidic soil)
- Estimated yield (metric tons/hectare), soil nitrogen & potassium, soil moisture, and optimal temperature range
- An AI advisory: plain-English and Pidgin explanations, a short list of actions, and one key takeaway

Example

Maize · Ogbomoso · pH 5.8 · low rainfall returns a 42% yield probability with High drought risk. The Pidgin advisory tells the farmer, in plain terms, to switch immediately to a drought-resistant hybrid seed ("Oba Super 6") — protecting the season’s investment before it is lost.

Core Architecture — Three Pillars

- **Cellular IoT Soil Probes —** low-cost, solar-powered telemetry measuring moisture, NPK, temperature, and rainfall in cooperative fields, built for Sahel and Savannah conditions.
- **Language Translation Pipeline —** an AI layer (Gemini) that maps numeric outputs onto natural, spoken advisory in local dialects, suitable for audio broadcast and IVR.
- **Cooperative Portfolio Node —** a dashboard where union administrators review regional alerts, dispatch fertilizer and seed, and manage operations across villages.

Target Users & Use Cases

Growa AI is a multi-sided platform — the same underlying data powers different workflows for each stakeholder in the agricultural value chain.

- **Smallholder farmers —** get personalized sowing, fertilizer, and risk advice via app, SMS, or a toll-free IVR line in their own dialect, paid for with small airtime micro-transactions.
- **Agrarian cooperatives —** plan sowing and fertilizer dispatch across village clusters from one dashboard. Trial cohorts show a 94% advisory sowing-compliance rate and a 3.1x increase in seasonal yield profit.
- **Extension agents —** equipped with portable soil scanners and offline-capable digital summaries for field visits in low-connectivity areas.
- **Microfinance & credit underwriters —** query real-time, hyper-local soil-risk and rainfall metrics to assess agricultural loan applications more accurately.
- **Crop insurance brokers —** subscribe to a Risk-API streaming live soil moisture and rainfall grids, improving pricing and payout accuracy of weather-indexed insurance.

Why Now

IoT sensor costs have fallen enough for field-level deployment at cooperative scale, while modern language models have, for the first time, made fluent generation of West African Pidgin and other local dialects practical. Growa AI sits at the intersection of both trends, applying inexpensive sensing and generative AI to a market mobile penetration has already reached — but technical agronomy content has not.

Market Opportunity

38M+ — TAM, Nigerian farmers | 3.2M — SOM, cooperative-union farmers | +18% — YoY growth in sowing-advice inquiries

Agriculture is over 22% of Nigeria's real GDP, with maize, yam, and cassava production alone valued at roughly $22B annually. Growa AI initially targets the 3.2M farmers organized under cooperative union frameworks — the segment best placed to adopt cohort-level sensing and advisory — before expanding to the broader 38M-farmer base. The model replicates across Ghana, Togo, Benin, and Côte d'Ivoire, which share comparable smallholder structures, crop mixes, and dialect-driven communication needs.

Business Model

- **B2C micro-transactions —** farmers pay roughly NGN 150 per advisory query, billed against mobile airtime balances, a model familiar from prepaid mobile services.
- **B2B enterprise SaaS —** cooperative administrators pay subscriptions for cohort dashboards, sowing-compliance tracking, and fertilizer-dispatch planning across member villages.
- **Risk-API subscriptions —** crop-insurance brokers and microfinance institutions pay for programmatic access to real-time local soil-moisture and rainfall data grids.

This three-layer model keeps per-farmer cost low enough for individual adoption while creating high-margin, recurring B2B and API revenue — projected to reach $1.4M ARR within 18 months of regional cohort scale-up.

Execution Roadmap

- **Adaptive voice (IVR) —** finalize an offline-capable, native-dialect text-to-speech engine for low-connectivity areas.
- **Distributed probes —** deploy solar-powered NPK and moisture sensor probes to partner cooperative fields across initial pilot regions.
- **Sowing onboarding —** train 500 rural extension leaders to manage multi-village cohorts alongside the app.

500+ — extension leaders (target) | 98% — dialect-IVR coverage (target)

Technology Stack

The current build is a React 19 + Vite single-page app styled with Tailwind CSS, with an interactive prediction demo, an investor pitch-deck view, and mocked auth/profile flows for cooperative and extension-agent personas. The prediction engine runs a deterministic agronomic model today; the production roadmap layers in Gemini-based generation for Pidgin and English advisory text, plus live ingestion from field IoT probes. The frontend deploys as a static site (e.g. Vercel), with sensor ingestion and AI-advisory generation handled by lightweight serverless API routes.

Vision

Growa AI's long-term goal is to make precision agronomy legible to every farmer in West Africa, regardless of literacy or language — turning sensor data that already exists into a daily, trusted voice in the field. Starting with cooperatives, where compliance and impact are directly measurable, Growa AI builds the data and trust needed to expand into credit, insurance, and pan-regional agricultural infrastructure.